

Answer Key & Marking Scheme

Subject: Science | **Test:** Test 2 (Day 14, 23 May 2026) — 2-Week Cumulative
Maximum Marks: 80

Syllabus: Chs 1–6: Chemical Reactions and Equations • Acids, Bases and Salts • Metals and Non-metals • Carbon and its Compounds • Life Processes • Control and Coordination

SECTION A — Answer Key (1 mark each)

- Q1. (b) decomposition.
- Q2. (c) 4.
- Q3. (b) zinc.
- Q4. (c) -COOH .
- Q5. (a) $\text{C}_2\text{H}_5\text{OH}$ (ethanol).
- Q6. (b) chloroplast.
- Q7. (c) 4.
- Q8. (a) neuron.
- Q9. (c) Cytokinin.
- Q10. (c) insulin.
- Q11. (a) Fluorine.
- Q12. (a) Cu and Sn.
- Q13. (b) double.
- Q14. (b) HCl.
- Q15. (b) transport of water and minerals.
- Q16. (a) cerebellum.
- Q17. Isomers are compounds with the same molecular formula but different structural formulae and properties.
- Q18. Carbon dioxide (CO_2).
- Q19. Haemoglobin is a red-coloured iron-containing pigment in RBCs that binds and transports oxygen from lungs to body tissues.
- Q20. CaOCl_2 (calcium oxychloride).

SECTION B — Answer Key (2 marks each)

- Q21. $2\text{Al} + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3 + \text{heat}$. The reaction releases so much heat that the iron formed is in molten state — used to weld broken railway tracks. Hence, thermite reaction. (2)
- Q22. (i) Carbon compounds (e.g. CH_4 , petrol, LPG) release a large amount of heat on combustion (high calorific value). (ii) They are easily transportable, can be stored, and burn cleanly under proper conditions. (2)
- Q23. Double circulation: blood passes through the heart twice in one complete cycle — once via pulmonary circulation (heart-lungs-heart) and once via systemic circulation (heart-body-heart). It keeps oxygenated and deoxygenated blood separate, ensuring efficient oxygen delivery — essential for warm-blooded animals with high energy needs. (2)
- Q24. (i) Aerobic uses O_2 ; anaerobic occurs without O_2 . (ii) Aerobic releases more energy and produces CO_2 + H_2O ; anaerobic produces lactic acid (in muscles) or ethanol + CO_2 (in yeast) and less energy. (2)
- Q25. Chemical messengers (neurotransmitters) are released only from the axonal end (presynaptic) and have receptors only at the dendritic end of the next neuron (postsynaptic). Hence, signals travel in only one direction. (2)
- Q26. NaHCO_3 on heating releases CO_2 which makes the dough rise/spongy. The acid (e.g. tartaric acid) neutralises the otherwise bitter sodium carbonate left behind, ensuring good taste. (2)

SECTION C — Answer Key (3 marks each)

Q27. Oxidation: gain of oxygen / loss of hydrogen / loss of electrons. Reduction: loss of oxygen / gain of hydrogen / gain of electrons. Redox reaction: simultaneous oxidation and reduction. Example: $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$. C is oxidised (gains O); ZnO is reduced (loses O). (3)

Q28. The electrolysis of brine (concentrated NaCl solution) produces NaOH (at cathode), Cl_2 (at anode) and H_2 . Equation: $2\text{NaCl}(\text{aq}) + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH}(\text{aq}) + \text{Cl}_2\uparrow + \text{H}_2\uparrow$. Uses: NaOH — soap, paper; Cl_2 — bleaching, water treatment, PVC; H_2 — fuel, ammonia synthesis. (3)

Q29. A soap molecule has two ends — a long hydrocarbon (hydrophobic) tail and an ionic (hydrophilic) head. In water, soap molecules form micelles around oil/grease — tails stuck inwards into the dirt, heads outward in water. The micelles are washed away with rinsing water, taking the dirt with them. (3)

Q30. Each kidney has about a million nephrons (the structural and functional unit). A nephron consists of a Bowman's capsule containing a tuft of capillaries called the glomerulus, followed by the tubule that has proximal and distal convoluted parts and the loop of Henle. Filtration occurs at the glomerulus (under pressure); useful substances (glucose, salts, water) are reabsorbed along the tubule; the filtrate that remains is urine, collected in the collecting duct. (Diagram with labels: Bowman's capsule, glomerulus, PCT, loop of Henle, DCT, collecting duct.) (3)

Q31. (i) **Thyroxine** — secreted by the thyroid gland; regulates carbohydrate, protein and fat metabolism. (ii) **Growth hormone** — secreted by the pituitary; controls growth of bones and muscles; deficiency causes dwarfism, excess causes gigantism. (iii) **Adrenaline** — secreted by the adrenal glands during stress; raises heart rate, breathing and supplies more energy to muscles ('fight or flight'). (3)

Q32. Metals high in the activity series have a greater tendency to lose electrons and form positive ions. They displace less reactive metals from their salt solutions. Example: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. Zn (more reactive) displaces Cu (less reactive). (3)

Q33. Saturated hydrocarbons (alkanes) have only single C–C bonds, while unsaturated hydrocarbons (alkenes/alkynes) have at least one double or triple C–C bond. Unsaturated compounds are more reactive because the multiple bonds are easily broken and they can undergo addition reactions (e.g. with H_2 , Br_2 , etc.). (3)

SECTION D — Answer Key (5 marks each)

Q34. (a) (i) Ethanol + Na: $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2\uparrow$ (sodium ethoxide and hydrogen). (ii) Hot conc. H_2SO_4 : $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_2=\text{CH}_2 + \text{H}_2\text{O}$ (dehydration to ethene). (3) (b) Advantages: cleaner burning (less CO/SOx); renewable (from sugar/grain). Disadvantage: lower energy content than petrol; food-vs-fuel concern with grain-based ethanol. (2)

Q35. (a) Diagram with labels: nostrils, nasal cavity, pharynx, larynx, trachea, bronchi (left and right), bronchioles, alveoli, diaphragm, lungs, ribs. (3) (b) Air enters alveoli; oxygen diffuses from alveoli (high partial pressure) into blood capillaries (low partial pressure) and binds with haemoglobin in RBCs to form oxyhaemoglobin. Carbon dioxide diffuses from blood (high) into the alveoli (low) and is exhaled out. Thin alveolar walls and large surface area facilitate efficient exchange. (2)

Q36. (a) **Reflex action** is a quick, involuntary response to a stimulus, controlled by the spinal cord. **Reflex arc:** Receptor → Sensory neuron → Spinal cord (interneuron) → Motor neuron → Effector (muscle/gland). Flow: stimulus → receptor → sensory neuron → spinal cord → motor neuron → effector → response. (3) (b) Nervous responses are faster as electrical impulses travel along nerves rapidly (e.g. withdrawing hand from hot object). Hormonal responses are slower as hormones are carried by blood and reach all body parts (e.g. growth, secondary sexual characteristics). Nervous: short-term, specific, fast; Hormonal: long-term, widespread, slower. (2)

SECTION E — Answer Key (4 marks each)

Q37. (i) Calcium sulphate hemihydrate, $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$. (1) (ii) On heating gypsum at ~373 K: $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + 1\frac{1}{2}\text{H}_2\text{O}$. (1) (iii) On adding water it sets into a hard solid (gypsum) due to rehydration. (1) (iv) Moisture would convert it back into gypsum, and it would no longer set when used. (1)

Q38. (i) A series of carbon compounds where successive members differ by $\text{--CH}_2\text{--}$ and have the same general formula and similar chemical properties. (1) (ii) C_4H_{10} . (1) (iii) Butane: $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--CH}_3$ (straight chain); 2-methylpropane: $(\text{CH}_3)_3\text{CH}$ (branched chain). (1) (iv) They share the same molecular formula but the spatial arrangement is different, leading to similar but not identical physical properties. (1)

Q39. (i) $6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{(sunlight, chlorophyll)}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. (1) (ii) Chlorophyll absorbs sunlight and provides the energy needed for the reaction. (1) (iii) Stomata allow exchange of gases — CO_2 enters the leaf, O_2 exits, and water vapour is lost (transpiration). (1) (iv) Light intensity, CO_2 concentration, temperature, water availability (any two). (1)

— End of Answer Key —